

EEG Report Classification — Full Results (MedGemma-27B)

Classifying free-text clinical EEG reports into five diagnostic categories with **MedGemma-27B** on CPU (llama.cpp, GBNF grammar-constrained, temperature 0). Ground truth is annotator **LD** (Reference Annotator); **SG** (Second Annotator) is the human ceiling; the paper’s **Mistral-7B** (Tian et al., Table III) is the external baseline.

We ran the **full 2×2×2 matrix** — everything is computed from results/*.json:

Axis	Options
Prompt	v1 (original, Impression-first) · v2 (professor’s: neurologist system role, body-first extraction, extended clinical definitions)
Dataset	Zoe (n=1495, in-distribution) · Maria (n=499, out-of-distribution — a different neurologist)
Quantization	Q2_K (~10 GB) · Q4_K_S (~15 GB)

Headline: on both datasets our small quantized model **matches or beats the paper’s Mistral-7B** and approaches the **human** ceiling on the main “abnormal vs normal” call. Q2_K is enough (Q4_K_S is no better). The professor’s v2 prompt is a **trade-off** — it helps overall abnormality (especially on Maria) but hurts the rare Focal Epi class.

Metric throughout: **Core F1** vs LD (binary present/absent, score ≥ 3 = present).

Headline — our best model vs Mistral-7B (all data pooled)

All data pooled (Zoe + Maria, n=1994) — our best vs Mistral-7B

Fully-correct reports (all 5 labels): ours 1672/1994 vs Mistral 1485/1994

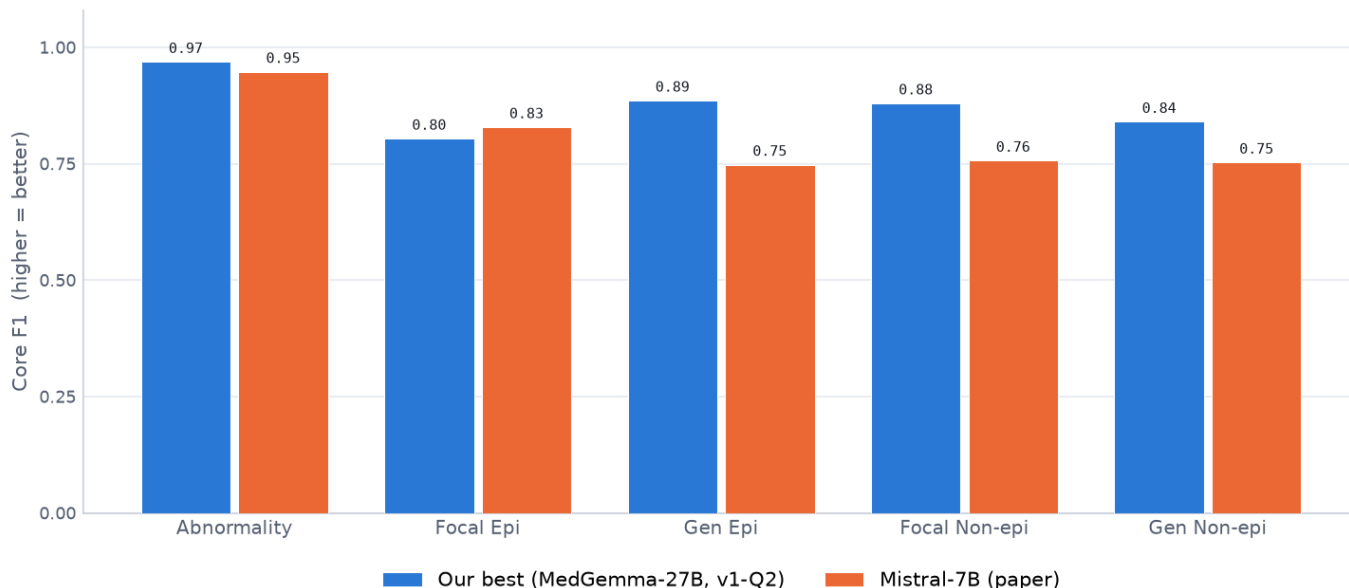


Figure 1: Pooled: our best vs Mistral-7B

What it shows: all **1994 reports** (1495 Zoe + 499 Maria) pooled into one set, comparing our best configuration (Q2_K, prompt v1) with the paper’s Mistral-7B. Both are scored against LD. **F1 is pooled correctly** — raw TP/FP/FN are summed across the two datasets and F1 is computed once (not an average)

of the two datasets' F1). Mistral's numbers use its **actual predictions** on the same 1994 reports (from the released classification database), verified to reproduce the paper's Table III per-dataset F1.

What you see: across all data our model **beats Mistral-7B on 4 of 5 categories** — decisively on the three harder ones (Gen Epi 0.89 vs 0.75, Focal Non-epi 0.88 vs 0.76, Gen Non-epi 0.84 vs 0.75) and slightly higher on Abnormality (0.97 vs 0.95). Mistral edges ahead only on the rare Focal Epi (0.83 vs 0.80). Overall, **1672 / 1994 reports fully correct (all 5 labels) vs Mistral's 1485** — a clear net improvement from a small, CPU-only, quantized model.

How exactly do we match the expert? — core vs exact level

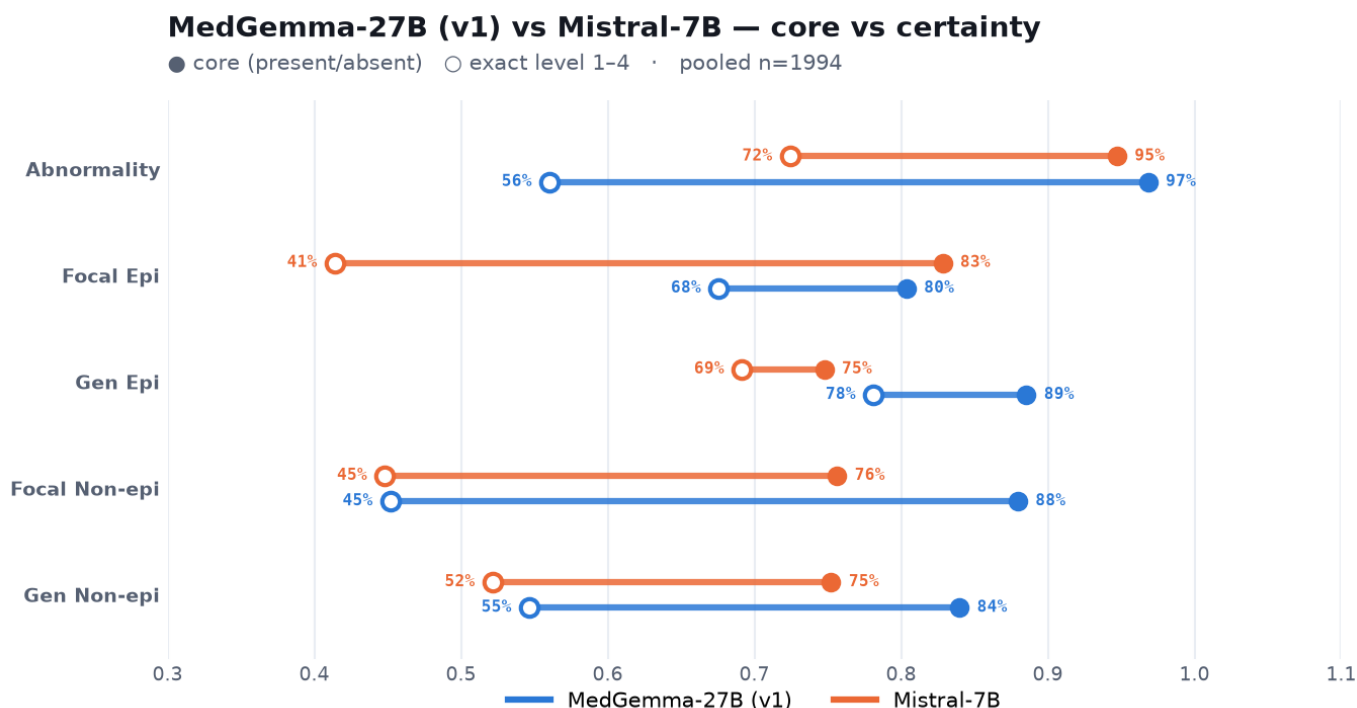


Figure 2: MedGemma v1 vs Mistral — core vs certainty

What it shows: MedGemma-27B (v1, blue) and Mistral-7B (orange) on one chart, all 1994 reports. Each line runs from **Core F1** (● — did we get present/absent right) to **Certainty F1** (○ — did we get the exact level 1/2/3/4). **The line length is the gap** — how much is lost when the exact confidence level is required.

What you see: on **core** both models sit far right; MedGemma is above Mistral on four of five categories. On the exact level the difference grows: on **Focal Epi** Mistral's line is huge (0.83 → **0.41**) — it gets the direction right but badly misses the level — while MedGemma barely drops (0.80 → 0.67). Same story on the non-epileptiform classes. The one place Mistral is better on the exact level is **Abnormality** (0.72 vs our 0.56) — judging *how sure* to be about “abnormal” is the hardest thing to imitate, exactly the gap the paper highlights.

Part 1 — Both datasets (Zoe + Maria)

1.1 Our model vs Mistral-7B vs human

What it shows: Core F1 per category for our Q2 model (v1), the paper's Mistral-7B, and the human second annotator — on each dataset.

Our model vs the paper's Mistral-7B and human — both datasets

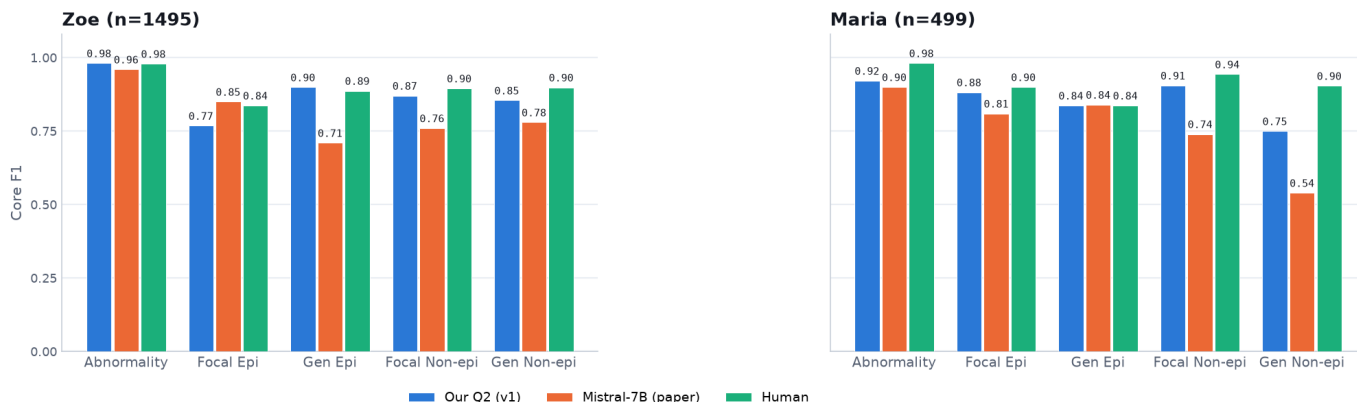


Figure 3: Core F1, both datasets

What you see: on **Zoe** we match/beat Mistral on 4/5 categories and equal the human on Abnormality (0.98). On **Maria** (a neurologist the model never saw) we still beat Mistral on 4/5 and stay close to human — the model generalizes rather than memorizing one reporting style.

1.2 Generalization: Zoe vs Maria

Generalization — same model (Q2 · v1) on both neurologists

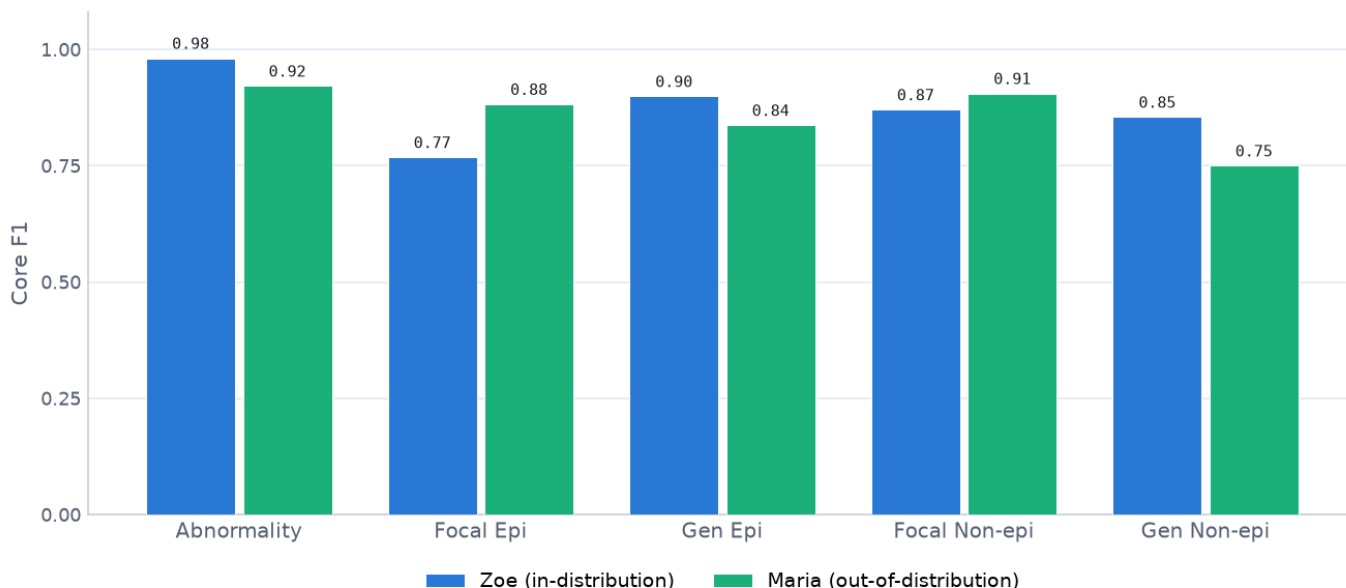


Figure 4: Generalization

What it shows: the same model (Q2 · v1) on both neurologists, side by side.

What you see: performance holds across styles. Abnormality drops a little on Maria (0.98 → 0.92) and Gen Non-epi is harder (0.85 → 0.75), but epileptiform and focal-non-epileptiform stay strong — no collapse on out-of-distribution data.

1.3 Confidence stays meaningful

What it shows: exact-match accuracy grouped by how confident the model is, on each dataset.

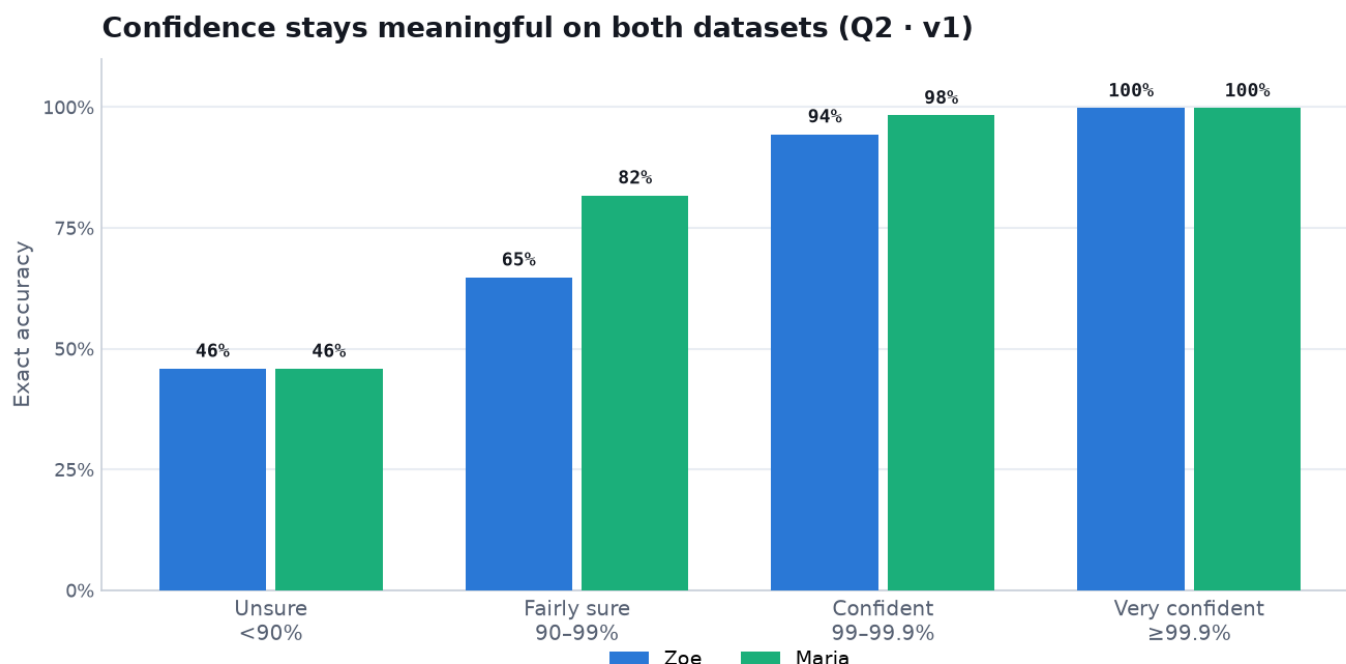


Figure 5: Reliability, both datasets

What you see: on both Zoe and Maria, accuracy climbs steeply with confidence — unsure predictions are right ~half the time, very confident ones ~100%. The model’s uncertainty is usable as a “route to a human” signal on new data too.

Part 2 — Zoe (n=1495, in-distribution)

2.1 Accuracy by category

What you see: Abnormality is at the human ceiling (0.98). The only real weak spot is **Focal Epi** (0.77) — a rare class where F1 is precision-limited. Prompt v2 (violet) lowers Focal Epi further (0.77 → 0.58); see 2.2.

2.2 Effect of the professor’s v2 prompt

What it shows: change in Core F1 from v1 to v2 (above 0 = v2 better).

What you see: on Zoe, v2 is mostly slightly negative, with a large drop on **Focal Epi**. Telling the model to read the report body first and not discard body findings makes it flag focal epileptiform activity more aggressively — more false alarms on an already precision-limited rare class. Fully-correct reports: v1 1255 → v2 1159.

2.3 Quantization: Q2_K vs Q4_K_S

What you see: the larger Q4_K_S is **not** better overall (and it’s slower to build/copy and ~50% larger). Q2_K is the right default.

2.4 Over- vs under-calling

What you see: on Zoe the model is **over-cautious** — 209 false alarms vs 98 misses (~2:1), concentrated in the non-epileptiform categories. Clinically the safer bias (an extra check beats a missed

Zoe (n=1495) — accuracy by category

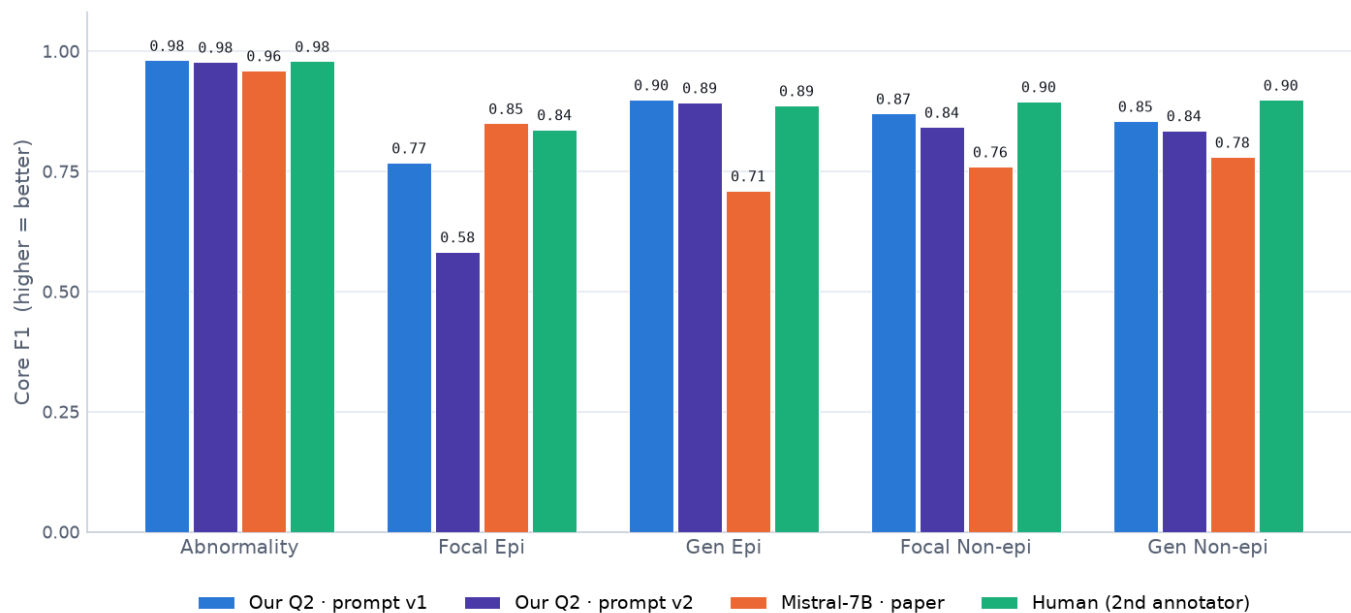


Figure 6: Zoe core F1

Zoe (n=1495) — effect of the professor's prompt (v2)

above 0 = v2 better · below 0 = v2 worse

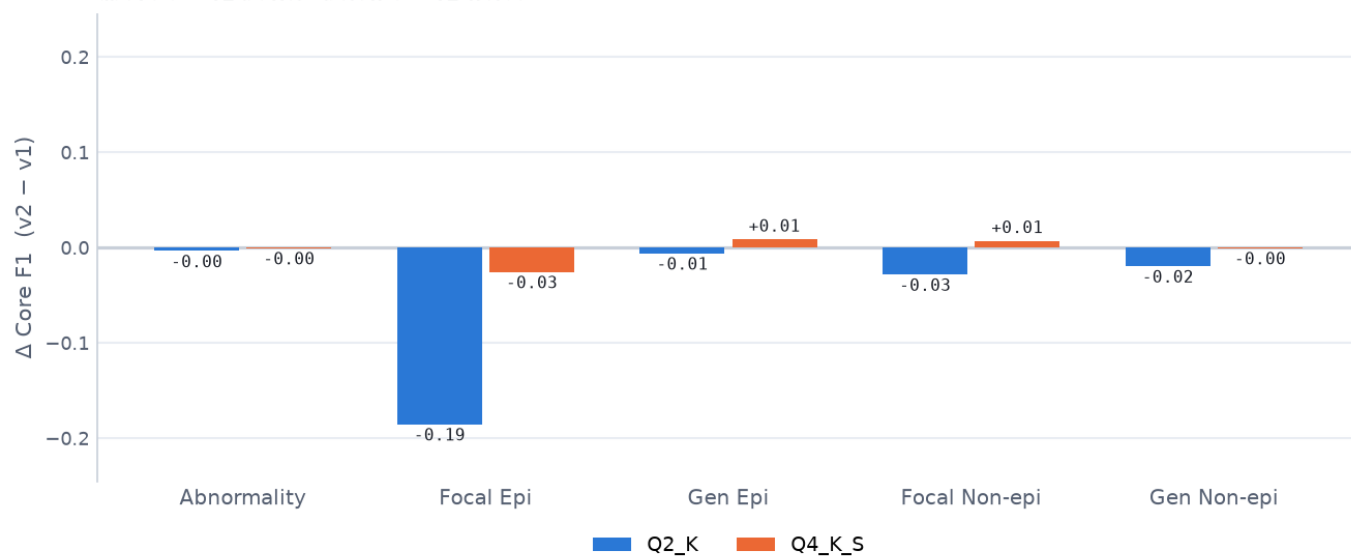


Figure 7: Zoe prompt effect

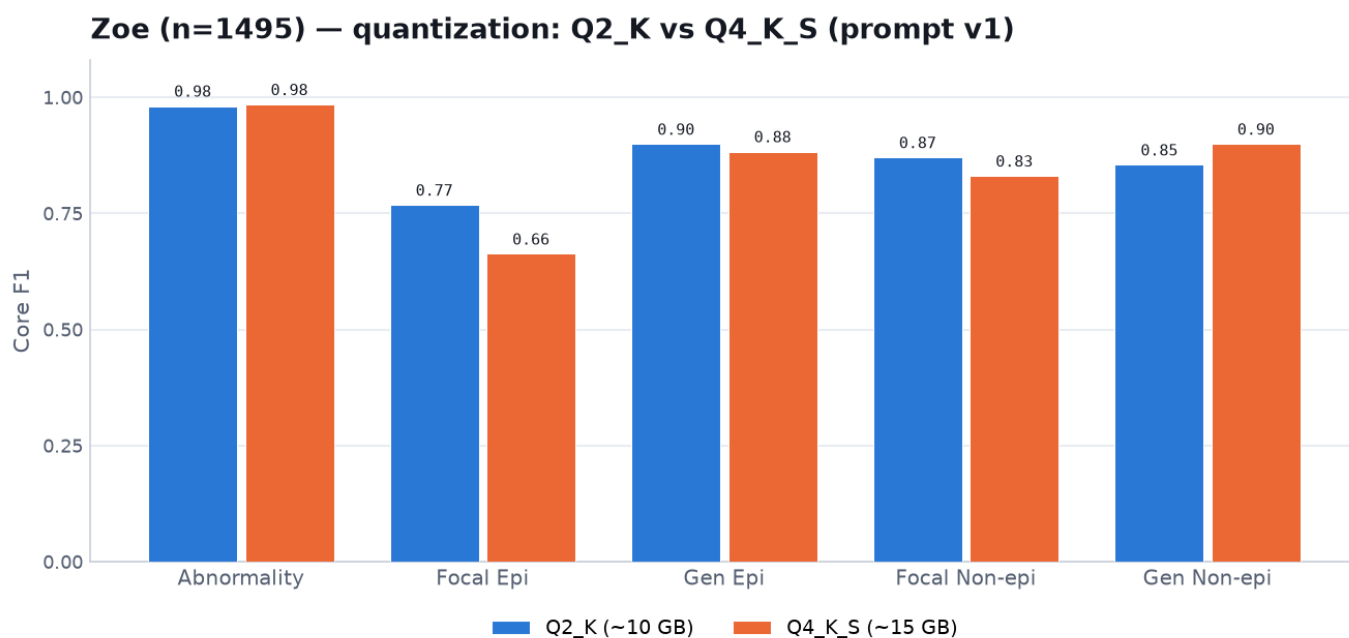


Figure 8: Zoe quant

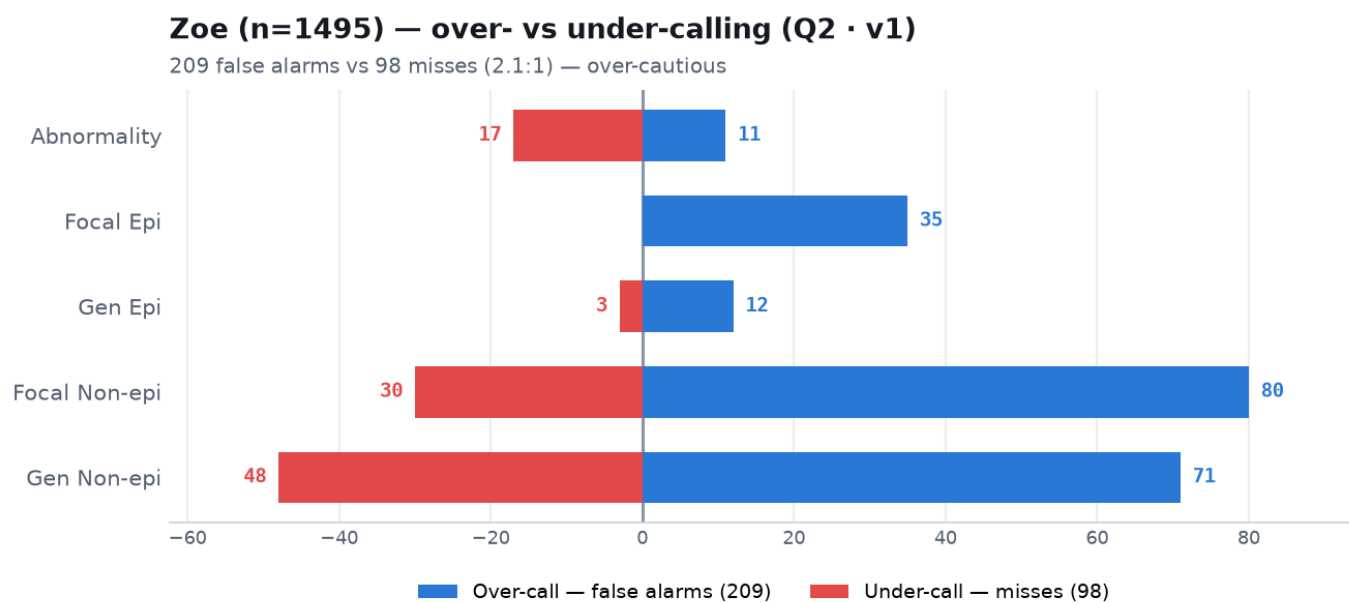


Figure 9: Zoe over/under

finding).

Part 3 — Maria (n=499, out-of-distribution)

3.1 Accuracy by category

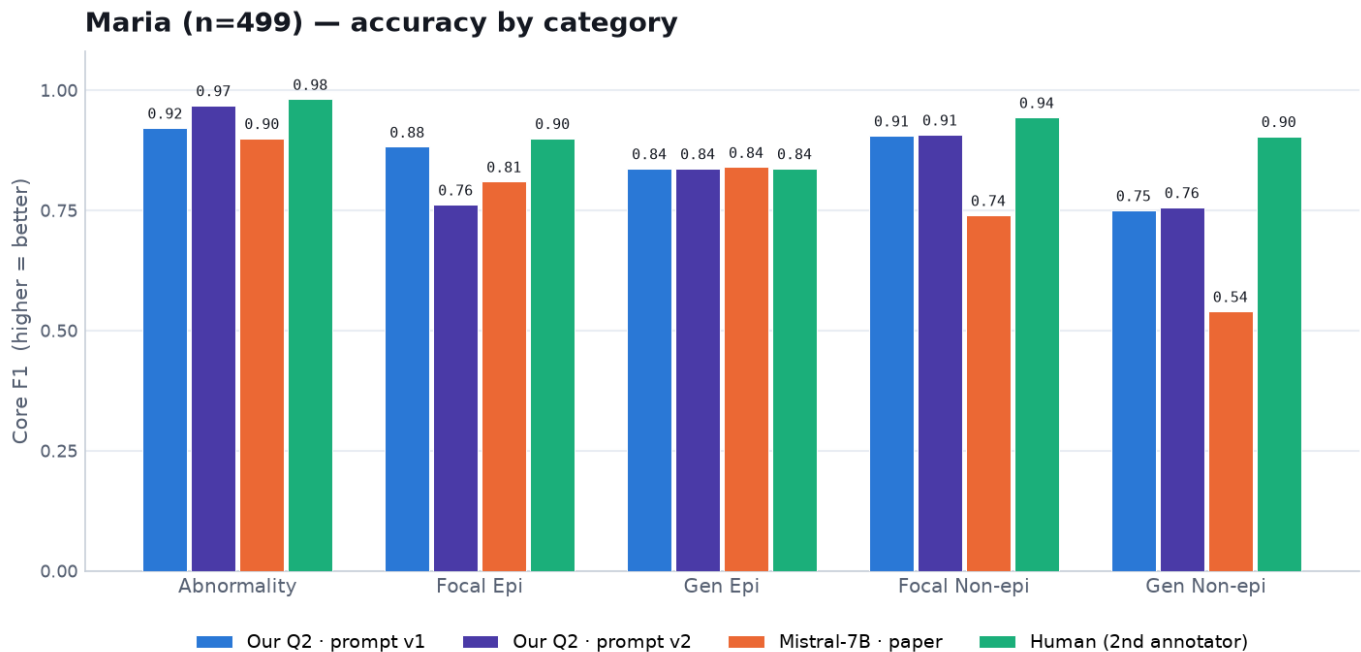


Figure 10: Maria core F1

What you see: the model holds up on an unseen neurologist — beating Mistral on 4/5. Abnormality 0.92 (Mistral 0.90; human 0.98). Here prompt **v2 helps** Abnormality (0.92 → 0.97, near human) — the opposite direction to Zoe.

3.2 Effect of the professor’s v2 prompt

What you see: on Maria, v2 **improves Abnormality (+0.05)** and most other categories, but again **hurts Focal Epi (−0.12)**. The best single config by fully-correct reports is **v2 + Q4_K_S (437/499)**.

3.3 Quantization: Q2_K vs Q4_K_S

What you see: on Maria, Q4_K_S is a bit better on the two hardest categories (Gen Non-epi 0.75 → 0.83), but the difference is small.

3.4 Over- vs under-calling

What you see: the key OOD difference — on Maria the model is **conservative** (38 false alarms vs 71 misses, 0.5:1), the **opposite** of Zoe. It misses more overall-abnormal (29) and diffuse-slowing (25) cases. This is exactly why v2 — which says “do not discard a body finding when the Impression is conservative” — raises Maria’s Abnormality: it converts some of those misses into correct calls.

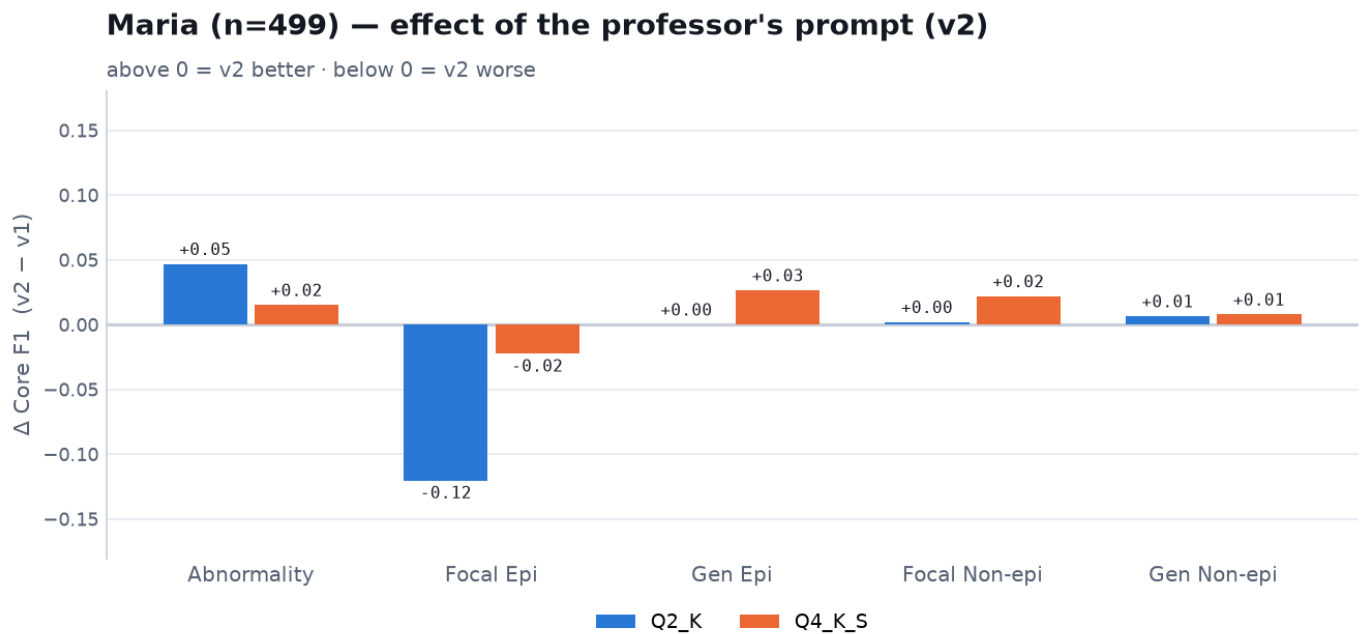


Figure 11: Maria prompt effect

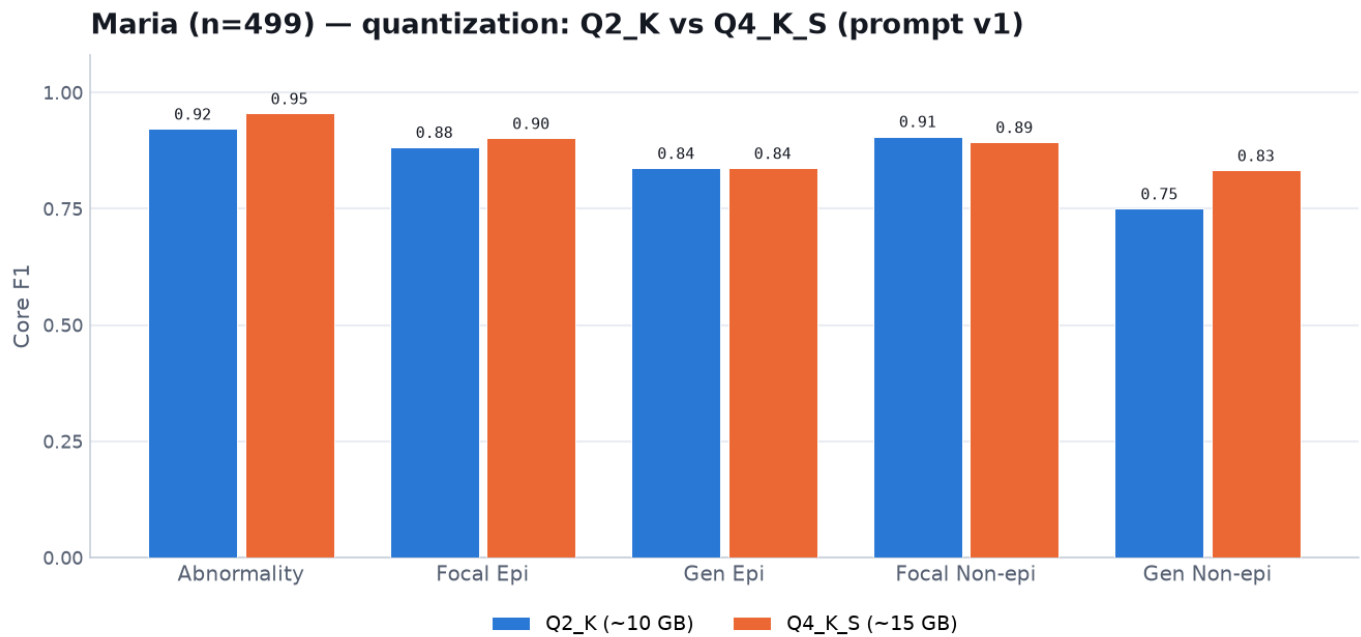


Figure 12: Maria quant

Maria (n=499) — over- vs under-calling (Q2 · v1)

38 false alarms vs 71 misses (0.5:1) — conservative

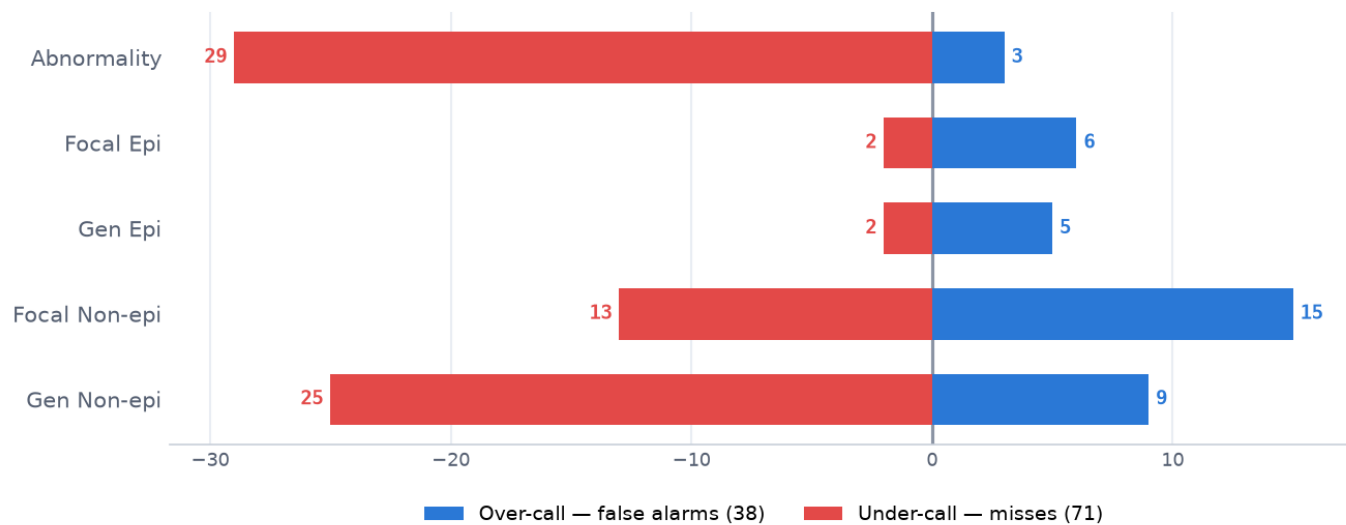


Figure 13: Maria over/under

Categories at a glance

Category	One-line meaning
Abnormality	Is the EEG abnormal at all.
Focal Epi	Epileptiform discharges (spikes) in one region .
Gen Epi	The same, but across the whole brain .
Focal Non-epi	Non-epileptic disturbance (slowing) in one region .
Gen Non-epi	Non-epileptic disturbance diffusely (e.g. encephalopathy).

Prompts

v1 (baseline) and v2 (professor's) prompts

Both live in [core/prompt.py](#), selected by the PROMPT_VARIANT env var. v1 uses a short definitions block and treats the Impression as the primary source; v2 moves the neurologist role into a system message, reads the report body first for the four subtypes, keeps body findings even when the Impression is conservative, and uses an extended ACNS/ILAE-based definitions block. Full text: `python -c "import core.prompt as p; print(p.PROMPT_PREFIX); print(p.PROMPT_PREFIX_V2)"`.

Reproduce

Every figure and number regenerates from the result JSONs:

```
source .venv/bin/activate
python -m analysis.plot_full           # all figures in this report

# a run (dataset / quant / prompt are env-selected):
DATASET=maria GGUF_QUANT=Q4_K_S PROMPT_VARIANT=v2 CTX_SIZE=8192 \
  sbatch cpu/run_benchmark.sbatch 0 499 results/out.json
```

Config	Result file
Zoe · v1 · Q2 / Q4	q2_cpu_full_n1495.json · cpu_q4_k_s_full_n1495.json
Zoe · v2 · Q2 / Q4	zoe_v2_cpu_q2_k_full_n1495.json · zoe_v2_cpu_q4_k_s_full_n1495.json
Maria · v1 · Q2 / Q4	maria_cpu_q2_k_full_n499.json · maria_cpu_q4_k_s_full_n499.json
Maria · v2 · Q2 / Q4	maria_v2_cpu_q2_k_full_n499.json · maria_v2_cpu_q4_k_s_full_n499.json

Model: MedGemma-27B GGUF (Q2_K / Q4_K_S), llama.cpp grammar-constrained, temp 0, 64-core CPU (rrg-rmcintos_cpu). Runtimes: Zoe ~4.5–5 h, Maria ~1.5 h per run.